

CS-013 Work Orders & Part Traceability

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Title	Work Orders & Part Traceability
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Supersedes	Master Tracker part rows as the only part record

Approvals

Role	Name	Date
Author	Simon Starbuck (drafted with Claude, SN6 Resources)	2026-07-12
Reviewer	simon (reviewer agent)	
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial issue, defines the WO record and its authority rules
B	2026-07-28	Claude	Corrected the app description throughout: it's the live Firebase app now, not <code>work-orders.html</code> ; buy-off, Backup/Restore, and photo-attachment wording updated to match. Language pass (fewer em dashes, less decorative bold).

1. Purpose

The Master Tracker did its job at the fleet level: parts, owners, deadlines, all visible. What it never gave us was the *part-level* record: steps, buy-offs, the stack as actually laid, measurements, photos. All of that lived in Slack threads and people's heads (PP-09). A work order (WO) is one part's complete record, spec through blockers through execution through evidence, on a screen or printed and kept with the part.

2. Scope

Every manufactured composites part and mold, including small cross-team jobs (the catch-can class of work, PP-05) and R&D coupons feeding decisions. If it consumes carbon or machine time, it gets a WO.

3. References

Ref	Document	Where
R1	PP-09, PP-05, PP-03 RCAs	FEB-COMP-PP-001
R2	The work-order app (live, Firebase-backed)	03 App/app/README.md, live at feb-composites.web.app
R2b	work-orders.html , the original offline single-file version — kept as a backup/archive viewer only, not the primary system	03 App/work-orders.html
R3	CS-002 (stack), CS-003 §7.2 (design review), CS-006/007 (process records)	this series
R4	CS-000 (revision rules apply to WOs)	CS-000

4. Definitions

- **WO ID** — W0-SNx-###, assigned from the app (or the next free number in the printed log). Never reused.
- **Buy-off** — the signed-in team member’s name, email, and timestamp, recorded by the app when they click a step’s buy-off button (self-buy-off allowed except where a step names an independent reviewer, e.g. mold design review). That’s a real step up from typed initials, but it isn’t tamper-proof: any roster member can still edit any record, so treat it as “who was signed in when this was clicked,” not a notarized signature.
- **Blocker step** — a step that blocks everything after it until it’s bought off (stack freeze, design review, drop test).
- **Retro WO** — a back-filled historical record (all W0-SN5-*); its unverifiable fields say “not recorded (retro)” — retro records are honest references, not exemplars of process compliance.

5. Equipment & Materials

The WO app (R2): a shared workspace at feb-composites.web.app, running on Firebase. Sign in with email and password; a lead has to add your email to the roster before you can see or touch anything. Every edit saves live to a shared database, so the whole team sees the same record at once; there’s nothing to import or merge. **work-orders.html** (R2b) still opens any exported JSON with no server at all, and stays useful as an offline archive viewer, but it’s not where day-to-day work happens anymore. Printed WOs: any printer + a pen at RFS.

6. Safety

No physical hazards; process standard. (Safety content lives in the process standards each WO step cites.)

7. Procedure

7.1 Create

1. Open the app → New Work Order → it assigns the next W0-SN6-###.
2. Fill at creation (this is the PP-04/PP-05 fix — requirements exist before work): part name, subteam, ME/RE, due date, process type, **stack (per CS-002)**, BOM, **qualityChecks with numeric targets**, and the standard step list for the process type (the app pre-fills steps from the templates; delete inapplicable ones consciously, don’t skip silently).
3. Cross-team parts: the requesting subteam’s requirements (thickness, DXF rev, mass) go in the WO and their lead is named on the release sign-off.

7.2 Execute

1. Work the steps in order; blocker steps block until bought off (the app enforces this; on paper the blocker rows are shaded: no initials, no moving on).
2. Each step: status, buy-off (signed-in name + timestamp), notes, photo references (phone photos → the part’s Drive folder, filename written on the WO step — WO steps don’t take a real upload yet; Tickets and Documents do, see 03 App/app/README.md).
3. Deviations from the frozen stack/spec = WO revision (CS-000 rules): quick, visible, counted. The revision letter rides the WO header.
4. Update `mold.location` whenever the mold physically moves (PP-10).

7.3 Close

1. Fill qualityChecks actuals (mass, thickness, Ω table, etc.), attach final photos, mark status Complete.
2. Nothing to export by hand: every save writes straight to the shared database and is live for the whole team immediately. The durable archive is the lead’s **monthly Backup** (header button) into the team Drive: Firestore is reliable, but a plain JSON file nobody can lock the team out of is the backup that matters. Restore, lead-only, reads that file back.
3. Print one copy for the physical binder at RFS if the part is safety/inspection-relevant (mech tech evidence).

7.4 Weekly use

1. Monday meeting: the app’s list view (filter: InWork) *is* the part status board — it replaces re-asking “where is X?” and feeds the weekly summary.
2. The lead runs Backup (one click, header) monthly — see §7.3. Nothing weekly is required since the database itself is always current; monthly is just the archive cadence.

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Every active part has a WO	100% of parts consuming carbon/machine time	Lead check at Monday meeting	—
Blocker steps bought off in order	No downstream work past an unfinished blocker	App enforcement / shaded rows on paper	WO
Monthly Backup run	Every month during build season	Lead	Drive folder timestamps
Retro honesty	No fabricated buy-offs on retro WOs	“not recorded (retro)” convention	WO fields

9. Records

The live database is the team’s day-to-day part-history record, shared and current for everyone at once; the monthly Backup JSON in Drive is the archive nobody can lock the team out of. Master-Tracker-style views (spreadsheets, summaries) may continue as *views* built from that record. The WO is the source of truth (same rule as CS-002 §9).